# vakint p-integrals: RustRed vs Forcer

2026-11-10 · itphlies · gammaloop 1a8862a97 · rustred b7fd9f2b · p-library-1-4 (external-data)

**Reading guide.** Each cell is the sum over all tops of that loop count at the given dots, times a numerator of that rank. Bold: RustRed cold (seconds, median). F: Forcer cold (Fw: warm, where only that was measured). ×: the ratio RustRed cold over Forcer; green is RustRed faster. ws, wc: RustRed warm on its own cache, and warm on one cache shared along the cells of that loop count. ✓: result exactly equal to Forcer's; n.c.: not exact-checked (cost; `p_exact.json`). n: integrals in the input. †: fewer repetitions than its run planned. **Sources:** 4L RustRed measured on the scope 1-4 library (p-library-1-4, external-data); every Forcer row (except Forcer 4L at 1 thread, cold, one repetition: measured here, v1 had none) and RustRed at 1L-3L cited from benchmark v1 (run `2026-11-08T18:42:02Z-84a7c0dee556`, gammaloop a659e3bf8, built-in 1-3 library). Identity caveat: Forcer, FORM and vakint's Forcer path are unchanged since v1, and the built-in 1-3 route is unchanged; Table 6 measures 1L-3L under the 1-4 program (one cell per loop count; the cross-program cost in full: M3-T19 S2). **Caveat:** indicative timings on a shared, loaded host. Benchmark v1 (cited rows): median foreign share of the measuring CPUs 1L n/a (before the sampler), 2L 0.82, 3L 0.12, 4L 0.52; v1's RustRed cold and wc include the after-evaluation cache save (fsync on the ZFS work dir): 122 of 624 saving children stalled 2-18 s, none of the non-saving children nor Forcer at 1L-3L. Measured rows save the cache after the clock (save seconds in Table 5 and the sub-lines). Measured rows (this run, window CPUs 128-143 of node 4): median foreign share of the measuring CPUs 1L 0.17, 2L 0.11, 3L 0.20, 4L 0.09 (4L: RustRed 1 thread 0.04, 16 threads 0.12, Forcer 1 thread 0.18); CPU 129 hosted two idle foreign pinned processes, and our own exact-tier writer ran on CPUs 148-151 of node 4 until 12:02Z, during the 1-thread 4L RustRed rows.

**Performance gates** (recorded, not blocking; the 4L target blocks in M5). 4L per cell (Table 5): on par with Forcer at 1 thread (≤ ×1.1): 8 of 16 met; better at 16 threads: 7 of 16 met; cold start dominated by loading: 1 of 10 met. Load of a fresh process: 5.15 s (median, p-library-1-4). G10b of v1, 3L rank ≥ 4, 16 threads, RustRed cold ≤ Forcer: ×0.14 to ×1.03, 9 of 9 cells.

RustRed faster ×1/4  ×4 Forcer faster

Table 1: 1 thread (`n_cores` 1, `form`); RustRed arithmetic `modular_polynomial`.

| L | dots | rank 0 | rank 2 | rank 4 | rank 6 | rank 8 |
|---|---|---|---|---|---|---|
| 1 L | 0 | **0.006** ✓<br>F 0.442 **×0.01**<br>ws 0.004 wc 0.006<br>n=1 | **0.007** ✓<br>F 0.436 **×0.02**<br>ws 0.004 wc 0.006<br>n=1 | **0.007** ✓<br>F 0.440 **×0.01**<br>ws 0.005 wc 0.007<br>n=1 | **0.007** ✓<br>F 0.436 **×0.02**<br>ws 0.004 wc 0.007<br>n=1 | **0.007** ✓<br>F 0.440 **×0.02**<br>ws 0.005 wc 0.007<br>n=1 |
| | 1 | **0.006** ✓<br>F 0.438 **×0.01**<br>ws 0.004 wc 0.007<br>n=1 | **0.008** ✓<br>F 0.435 **×0.02**<br>ws 0.005 wc 0.007<br>n=1 | **0.009** ✓<br>F 0.438 **×0.02**<br>ws 0.007 wc 0.008<br>n=1 | **0.011** ✓<br>F 0.444 **×0.02**<br>ws 0.008 wc 0.010<br>n=1 | **0.014** ✓<br>F 0.450 **×0.03**<br>ws 0.011 wc 0.013<br>n=1 |
| | 2 | **0.006** ✓<br>F 0.434 **×0.01**<br>ws 0.005 wc 0.006<br>n=1 | **0.006** ✓<br>F 0.436 **×0.01**<br>ws 0.005 wc 0.007<br>n=1 | **0.007** ✓<br>F 0.437 **×0.02**<br>ws 0.005 wc 0.006<br>n=1 | **0.007** ✓<br>F 0.441 **×0.02**<br>ws 0.005 wc 0.006<br>n=1 | **0.007** ✓<br>F 0.441 **×0.02**<br>ws 0.005 wc 0.006<br>n=1 |
| 2 L | 0 | **0.195** ✓<br>F 0.482 **×0.41**<br>ws 0.006 wc 0.013<br>n=1 | **8.43** ✓<br>F 0.493 **×17**<br>ws 0.005 wc 0.196<br>n=1 | **0.392** ✓<br>F 0.478 **×0.82**<br>ws 0.006 wc 0.128<br>n=1 | **0.203** ✓<br>F 0.489 **×0.42**<br>ws 0.007 wc 0.286<br>n=1 | **0.150** ✓<br>F 0.526 **×0.28**<br>ws 0.008 wc 0.709<br>n=1 |
| | 1 | **0.013** ✓<br>F 0.483 **×0.03**<br>ws 0.006 wc 0.067<br>n=1 | **0.126** ✓<br>F 0.501 **×0.25**<br>ws 0.006 wc 0.363<br>n=1 | **0.078** ✓<br>F 0.498 **×0.16**<br>ws 0.006 wc 0.676<br>n=1 | **0.075** ✓<br>F 0.488 **×0.15**<br>ws 0.007 wc 0.549<br>n=1 | **0.192** ✓<br>F 0.517 **×0.37**<br>ws 0.008 wc 0.406<br>n=1 |
| | 2 | **0.440** ✓<br>F 0.484 **×0.91**<br>ws 0.007 wc 6.33<br>n=1 | **0.393** ✓<br>F 0.494 **×0.80**<br>ws 0.007 wc 0.520<br>n=1 | **2.68** ✓<br>F 0.501 **×5.36**<br>ws 0.007 wc 0.192<br>n=1 | **0.023** ✓<br>F 0.504 **×0.04**<br>ws 0.009 wc 0.536<br>n=1 | **0.026** ✓<br>F 0.515 **×0.05**<br>ws 0.011 wc 4.33<br>n=1 |
| 3 L | 0 | **0.566** ✓<br>F 1.66 **×0.34**<br>ws 0.021 wc 0.076<br>n=3 | **2.16** ✓<br>F 1.76 **×1.23**<br>ws 0.028 wc 1.44<br>n=3 | **0.286** ✓<br>F 2.01 **×0.14**<br>ws 0.018 wc 0.288<br>n=3 | **9.19** ✓<br>F 2.38 **×3.87**<br>ws 0.079 wc 0.587<br>n=3 | **1.04** ✓<br>F 2.82 **×0.37**<br>ws 0.228 wc 0.986<br>n=3 |
| | 1 | **3.48** ✓<br>F 1.93 **×1.80**<br>ws 0.023 wc 0.090<br>n=3 | **0.206** ✓<br>F 1.86 **×0.11**<br>ws 0.030 wc 0.399<br>n=3 | **1.37** ✓<br>F 2.02 **×0.67**<br>ws 0.031 wc 0.784<br>n=3 | **0.832** ✓<br>F 2.42 **×0.34**<br>ws 0.134 wc 0.831<br>n=3 | **1.60** ✓<br>F 2.88 **×0.56**<br>ws 0.458 wc 1.40<br>n=3 |
| | 2 | **0.345** ✓<br>F 1.96 **×0.18**<br>ws 0.026 wc 0.135<br>n=3 | **0.345** ✓<br>F 2.06 **×0.17**<br>ws 0.036 wc 0.234<br>n=3 | **1.47** ✓<br>F 2.21 **×0.66**<br>ws 0.035 wc 0.344<br>n=3 | **1.45** ✓<br>F 2.59 **×0.56**<br>ws 0.123 wc 0.888<br>n=3 | **1.28** ✓<br>F 3.32 **×0.38**<br>ws 0.385 wc 1.12<br>n=3 |
| 4 L | 0 | **4.58** ✓<br>F 19.7 **×0.23**<br>ws 1.68 wc 4.61<br>n=11 | **25.8** ✓<br>F 34.8 **×0.74**<br>ws 8.65 wc 25.5<br>n=11 | **92.8** n.c.<br>F 44.6 **×2.08**<br>ws 47.6 wc 88.8<br>n=11 | **> 10 min** n.c.<br>F 83.4 **> ×7.20**<br>n=11 | **> 10 min** n.c.<br>F 123 **> ×4.88**<br>n=11 |
| | 1 | **8.55** ✓<br>F 34.1 **×0.25**<br>ws 2.99 wc –<br>n=11 | **30.2** ✓<br>F 41.3 **×0.73**<br>ws 11.6 wc –<br>n=11 | **96.0** n.c.<br>F 47.5 **×2.02**<br>ws 49.6 wc –<br>n=11 | **> 10 min** n.c.<br>F 87.6 **> ×6.85**<br>n=11 | **> 10 min** n.c.<br>F 133 **> ×4.51**<br>n=11 |
| | 2 | **9.60** ✓<br>F 42.0 **×0.23**<br>ws 3.98 wc –<br>n=11 | **13.4** ✓<br>F 46.4 **×0.29**<br>ws 4.80 wc –<br>n=11 | **30.3** n.c.<br>F 55.9 **×0.54**<br>ws 9.31 wc –<br>n=11 | **> 10 min** n.c.<br>F 118 **> ×5.09**<br>n=11 | **> 10 min** n.c.<br>F 182 **> ×3.29**<br>n=11 |
| phys. | | 2 L **0.357** ✓<br>F 0.519 **×0.69**<br>ws 0.022 wc 0.023<br>n=11 | 3 L **0.712** ✓<br>F 1.34 **×0.53**<br>ws 0.402 wc 0.567<br>n=53 | 4 L **87.4** ✓<br>F 119 **×0.73**<br>ws 78.6 wc –<br>n=293 | 5L canary: n/a | |

Table 2: 16 threads (`n_cores` 16, `tform -w16`); RustRed arithmetic `modular_polynomial`.

| L | dots | rank 0 | rank 2 | rank 4 | rank 6 | rank 8 |
|---|---|---|---|---|---|---|
| 1 L | 0 | **0.010** ✓<br>F 0.467 **×0.02**<br>ws 0.007 wc 0.012<br>n=1 | **0.012** ✓<br>F 0.458 **×0.03**<br>ws 0.007 wc 0.011<br>n=1 | **0.013** ✓<br>F 0.462 **×0.03**<br>ws 0.007 wc 0.011<br>n=1 | **0.014** ✓<br>F 0.475 **×0.03**<br>ws 0.008 wc 0.022<br>n=1 | **0.013** ✓<br>F 0.494 **×0.03**<br>ws 0.007 wc 0.015<br>n=1 |
| | 1 | **0.015** ✓<br>F 0.467 **×0.03**<br>ws 0.006 wc 0.043<br>n=1 | **3.97** ✓<br>F 0.454 **×8.76**<br>ws 0.015 wc 0.019<br>n=1 | **0.033** ✓<br>F 0.479 **×0.07**<br>ws 0.020 wc 0.023<br>n=1 | **0.038** ✓<br>F 0.486 **×0.08**<br>ws 0.026 wc 0.029<br>n=1 | **0.045** ✓<br>F 0.500 **×0.09**<br>ws 0.033 wc 0.036<br>n=1 |
| | 2 | **0.009** ✓<br>F 0.473 **×0.02**<br>ws 0.007 wc 0.012<br>n=1 | **0.666** ✓<br>F 0.460 **×1.45**<br>ws 0.008 wc 0.016<br>n=1 | **0.017** ✓<br>F 0.468 **×0.04**<br>ws 0.007 wc 1.08<br>n=1 | **0.012** ✓<br>F 0.459 **×0.03**<br>ws 0.008 wc 0.010<br>n=1 | **0.016** ✓<br>F 0.479 **×0.03**<br>ws 0.009 wc 0.010<br>n=1 |
| 2 L | 0 | **0.025** ✓<br>F 0.671 **×0.04**<br>ws 0.011 wc 0.027<br>n=1 | **0.845** ✓<br>F 0.594 **×1.42**<br>ws 0.016 wc 0.521<br>n=1 | **0.745** ✓<br>F 0.590 **×1.26**<br>ws 0.010 wc 0.024<br>n=1 | **0.579** ✓<br>F 0.630 **×0.92**<br>ws 0.027 wc 0.757<br>n=1 | **0.636** ✓<br>F 0.599 **×1.06**<br>ws 0.012 wc 0.400<br>n=1 |
| | 1 | **0.128** ✓<br>F 0.547 **×0.23**<br>ws 0.013 wc 0.242<br>n=1 | **0.021** ✓<br>F 0.575 **×0.04**<br>ws 0.012 wc 10.0<br>n=1 | **1.03** ✓<br>F 0.567 **×1.82**<br>ws 0.011 wc 0.146<br>n=1 | **0.149** ✓<br>F 0.576 **×0.26**<br>ws 0.010 wc 9.06<br>n=1 | **0.531** ✓<br>F 0.579 **×0.92**<br>ws 0.012 wc 0.570<br>n=1 |
| | 2 | **0.102** ✓<br>F 0.487 **×0.21**<br>ws 0.011 wc 0.021<br>n=1 | **0.025** ✓<br>F 0.611 **×0.04**<br>ws 0.010 wc 0.872<br>n=1 | **0.023** ✓<br>F 0.608 **×0.04**<br>ws 0.013 wc 0.219<br>n=1 | **0.035** ✓<br>F 0.547 **×0.06**<br>ws 0.013 wc 5.07<br>n=1 | **0.615** ✓<br>F 0.577 **×1.07**<br>ws 0.017 wc 0.634<br>n=1 |
| 3 L | 0 | **0.127** ✓<br>F 1.80 **×0.07**<br>ws 0.024 wc 0.215<br>n=3 | **0.630** ✓<br>F 1.95 **×0.32**<br>ws 0.034 wc 0.505<br>n=3 | **0.299** ✓<br>F 2.16 **×0.14**<br>ws 0.022 wc 2.83<br>n=3 | **0.523** ✓<br>F 2.50 **×0.21**<br>ws 0.097 wc 0.349<br>n=3 | **0.653** ✓<br>F 2.77 **×0.24**<br>ws 0.249 wc 0.847<br>n=3 |
| | 1 | **0.061** ✓<br>F 2.08 **×0.03**<br>ws 0.028 wc 0.065<br>n=3 | **0.455** ✓<br>F 1.99 **×0.23**<br>ws 0.038 wc 0.665<br>n=3 | **1.03** ✓<br>F 2.18 **×0.47**<br>ws 0.036 wc 0.134<br>n=3 | **0.476** ✓<br>F 2.55 **×0.19**<br>ws 0.161 wc 0.496<br>n=3 | **1.08** ✓<br>F 2.88 **×0.37**<br>ws 0.466 wc 1.03<br>n=3 |
| | 2 | **0.070** ✓<br>F 2.11 **×0.03**<br>ws 0.032 wc 6.58<br>n=3 | **0.409** ✓<br>F 2.22 **×0.18**<br>ws 0.045 wc 0.253<br>n=3 | **2.45** ✓<br>F 2.37 **×1.03**<br>ws 0.046 wc 5.66<br>n=3 | **0.424** ✓<br>F 2.67 **×0.16**<br>ws 0.164 wc 0.621<br>n=3 | **0.908** ✓<br>F 3.06 **×0.30**<br>ws 0.438 wc 5.06<br>n=3 |
| 4 L | 0 | **2.75** ✓<br>Fw 21.4 **×0.13**<br>ws 1.74 wc 2.68<br>n=11 | **14.7** ✓<br>Fw 36.9 **×0.40**<br>ws 6.81 wc 14.1<br>n=11 | **52.7** n.c.<br>Fw 47.5 **×1.11**<br>ws 33.8 wc 49.2<br>n=11 | **> 10 min** n.c.<br>Fw 70.1 **> ×8.55**<br>n=11 | **> 10 min** n.c.<br>Fw 95.3 **> ×6.30**<br>n=11 |
| | 1 | **4.60** ✓<br>Fw 38.6 **×0.12**<br>ws 2.54 wc —<br>n=11 | **17.4** ✓<br>Fw 45.0 **×0.39**<br>ws 9.18 wc —<br>n=11 | **53.4** n.c.<br>Fw 52.4 **×1.02**<br>ws 32.2 wc —<br>n=11 | **399** n.c.<br>Fw 76.6 **×5.21**<br>ws 247 wc —<br>n=11 | **> 10 min** n.c.<br>Fw 98.0 **> ×6.12**<br>n=11 |
| | 2 | **5.47** ✓<br>Fw 45.4 **×0.12**<br>ws 3.31 wc —<br>n=11 | **6.81** ✓<br>Fw 51.9 **×0.13**<br>ws 3.78 wc —<br>n=11 | **15.4** n.c.<br>Fw 65.2 **×0.24**<br>ws 7.24 wc —<br>n=11 | **273** n.c.<br>Fw 100 **×2.72**<br>ws 172 wc —<br>n=11 | **> 10 min** n.c.<br>Fw 132 **> ×4.53**<br>n=11 |
| **phys.** | | 2 L **0.175** ✓<br>F 0.628 **×0.28**<br>ws 0.062 wc 0.070<br>n=11 | 3 L **0.779** ✓<br>F 1.24 **×0.63**<br>ws 0.612 wc 0.766<br>n=53 | 4 L **84.4** ✓<br>Fw 65.4 **×1.29**<br>ws 78.2 wc —<br>n=293 | 5L canary: n/a | |

Table 3: RustRed cold, arithmetic `exact` against `modular_polynomial` (mod.), reduction-only seconds (median); gain = exact / mod. (1L-3L: benchmark v1)

| cell | 1 thread | | | 16 threads | | |
|---|---|---|---|---|---|---|
| | exact | mod. | gain | exact | mod. | gain |
| L1 d0 r0 | < 0.001 | < 0.001 | **×0.27** | 0.001 | 0.003 | **×0.30** |
| L1 d1 r4 | < 0.001 | 0.002 | **×0.27** | 0.005 | 0.025 | **×0.21** |
| L1 d2 r8 | < 0.001 | < 0.001 | **×0.43** | 0.002 | 0.008 | **×0.25** |
| L2 d0 r0 | 0.003 | 0.004 | **×0.76** | 0.003 | 0.009 | **×0.35** |
| L2 d1 r4 | 0.005 | 0.006 | **×0.84** | 0.005 | 0.011 | **×0.47** |
| L2 d2 r8 | 0.009 | 0.010 | **×0.93** | 0.010 | 0.026 | **×0.37** |
| physics L2 | 0.008 | 0.014 | **×0.61** | 0.020 | 0.077 | **×0.26** |
| L3 d0 r0 | 0.057 | 0.057 | **×1.00** | 0.029 | 0.032 | **×0.93** |
| L3 d1 r4 | 0.272 | 0.266 | **×1.02** | 0.109 | 0.105 | **×1.03** |
| L3 d2 r8 | 1.86 | 1.09 | **×1.70** | 0.860 | 0.733 | **×1.17** |
| physics L3 | 0.396 | 0.377 | **×1.05** | 0.408 | 0.422 | **×0.97** |

Table 4: Reuse, `modular_polynomial`: warm time over cold time per loop count (sums over the cells). ws clean: warm-same children with 0 misses and 0 inserts. wc: the chain's cache counters at 16 threads. (1L-3L: benchmark v1) The 1L-3L ratios are dominated by the cache-save stalls of v1's cold runs (caveat).

| L | 1 thread | | 16 threads | | ws clean | wc hits / misses |
|---|---|---|---|---|---|---|
| | ws/cold | wc/cold | ws/cold | wc/cold | | |
| 1 L | ×0.70 | ×0.93 | ×0.04 | ×0.28 | 90 / 90 | 336 / 67 |
| 2 L | ×1/107 | ×1.11 | ×0.05 | ×5.06 | 96 / 96 | 1207 / 1042 |
| 3 L | ×0.08 | ×0.39 | ×0.24 | ×2.52 | 96 / 96 | 78418 / 51148 |
| 4 L | ×0.55 | ×0.97 | ×0.64 | ×0.94 | 22 / 22 | 333952 / 1139864 |

Table 5: 4L speed target per cell (G10b; recorded, not blocking): RustRed cold over Forcer (measured at 1 thread, cited at 16 threads) at 1 and 16 threads; load of a fresh process against the one-off cost of a cold evaluation (cold − ws); for a missed thread count the breakdown of the cold run. Inside the reduction about a third is step generation and two thirds the ordered weight elimination (M3-T19 S2, 4 threads).

| cell | 1 thread (≤ ×1.1) | 16 threads (< ×1) | load / cold − ws, 1 thread | misses: threads: load / reduction / rest / save (s, RustRed cold) |
|---|---|---|---|---|
| L4 d0 r0 | ×0.23 **met** | ×0.13 **met** | 5.15 / 2.90 **met** | |
| L4 d0 r2 | ×0.74 **met** | ×0.40 **met** | 5.13 / 17.2 **missed** | |
| L4 d0 r4 | ×2.08 **missed** | ×1.11 **missed** | 5.16 / 45.2 **missed** | 1: 5.16 / 92.7 / 0.076 / 3.84<br>16: 5.13 / 52.7 / 0.081 / 3.95 |
| L4 d0 r6 | > ×7.20 **missed** | > ×8.55 **missed** | — / — — | 1: capped (> 10 min), no breakdown<br>16: capped (> 10 min), no breakdown |
| L4 d0 r8 | > ×4.88 **missed** | > ×6.30 **missed** | — / — — | 1: capped (> 10 min), no breakdown<br>16: capped (> 10 min), no breakdown |
| L4 d1 r0 | ×0.25 **met** | ×0.12 **met** | 5.18 / 5.56 **missed** | |
| L4 d1 r2 | ×0.73 **met** | ×0.39 **met** | 5.20 / 18.6 **missed** | |
| L4 d1 r4 | ×2.02 **missed** | ×1.02 **missed** | 5.48 / 46.4 **missed** | 1: 5.48 / 95.9 / 0.109 / 3.84<br>16: 5.16 / 53.3 / 0.126 / 3.72 |
| L4 d1 r6 | > ×6.85 **missed** | ×5.21 **missed** | — / — — | 1: capped (> 10 min), no breakdown<br>16: 5.18 / 399 / 0.366 / 21.4 |
| L4 d1 r8 | > ×4.51 **missed** | > ×6.12 **missed** | — / — — | 1: capped (> 10 min), no breakdown<br>16: capped (> 10 min), no breakdown |
| L4 d2 r0 | ×0.23 **met** | ×0.12 **met** | 5.11 / 5.62 **missed** | |
| L4 d2 r2 | ×0.29 **met** | ×0.13 **met** | 5.37 / 8.60 **missed** | |
| L4 d2 r4 | ×0.54 **met** | ×0.24 **met** | 5.09 / 21.0 **missed** | |
| L4 d2 r6 | > ×5.09 **missed** | ×2.72 **missed** | — / — — | 1: capped (> 10 min), no breakdown<br>16: 5.14 / 272 / 0.214 / 15.5 |
| L4 d2 r8 | > ×3.29 **missed** | > ×4.53 **missed** | — / — — | 1: capped (> 10 min), no breakdown<br>16: capped (> 10 min), no breakdown |
| physics L4 | ×0.73 **met** | ×1.29 **missed** | 5.23 / 8.80 **missed** | 16: 5.26 / 31.1 / 53.3 / 0.325 |

Table 6: 1L-3L under the scope 1-4 program (p-library-1-4), 1 thread, against the built-in 1-3 program of the cited rows; seconds (median). The 1-4 load is the price of one program for every loop count (M3 decision D-M3-1).

| cell | evaluate | | | reduction only | | load | |
|---|---|---|---|---|---|---|---|
| | 1-4 cold | 1-4 ws | 1-3 cold | 1-4 | 1-3 | 1-4 | 1-3 |
| L1 d1 r4 | 0.007 | 0.007 | 0.009 | 0.002 | 0.002 | 5.12 | 0.997 |
| L2 d1 r4 | 0.011 | 0.006 | 0.078 | 0.006 | 0.006 | 5.14 | 1.05 |
| L3 d1 r4 | 0.270 | 0.033 | 1.37 | 0.260 | 0.266 | 5.15 | 0.976 |

**Sub-lines** (seconds, sum over the cells measured at 1 thread).

| | 1 L | 2 L | 3 L | 4 L |
|---|---|---|---|---|
| RustRed reduction only | 0.016 | 0.118 | 7.14 | 345 |
| RustRed load (excluded) | 15.1 | 17.0 | 15.6 | 52.1 |
| RustRed cache save (after the clock) | — | — | — | 13.5 |
| Forcer FORM subprocess | 6.18 | 7.50 | 34.4 | 1176 |
| Raw FORM reference | 6.36 | 7.94 | 34.7 | — |

**Method.** Timed: the wall time of `Vakint::evaluate` in Retain mode (exact reduction onto masters, same frontend and finalizer for both backends) in one fresh process per measurement, with the input parsed, the engine and library loaded and the 1L bubble with four dots (in no cell: vakint memoises Forcer's FORM runs per process) evaluated once beforehand, by either backend, on directories of its own; repetitions: 1 or 3 (as each run planned; 4L: one), median. RustRed cold: empty ordered-cache directory, shipped caches off; the cache is saved after the clock. ws: a fresh process on a read-only copy of the cache of its cold run. wc: per loop count, one chain over the cells in table order sharing one cache from empty, each cell measured once in the chain. Forcer cold: private empty materialisation directory; Forcer warm: materialised. Forcer has no result cache. The raw reference runs the generated FORM programs directly, one process per top. OS pages are not flushed. Cap per measurement: > 10 min, > 30 min, > 60 min (by run; a capped cell shows the cap, its ratio the bound cap / Forcer). Cited 4L Forcer rows: 16 threads, warm, one repetition (benchmark v1). Physics row: massless-quark (L-1)-rung gluon self-energy, Feynman gauge, transverse projector, external momentum kept.

**Provenance.** Run `2026-11-10T11:27:54Z-37801b735425`, profile release, `FORM 5.0.0 (Jan  1 1980)`, Forcer 68a30bf, AMD EPYC 9754 128-Core Processor on itphlies, CPUs 1 thread: 128; 16 thread: 128-143, binding `--cpunodebind=4 --membind=4`, RustRed arithmetic `modular_polynomial` in Tables 1, 2, 4-6, `exact` in Table 3 (each child records the arithmetic it resolved), `RUSTRED_ORDERED_ADJOINT_TAIL` unset, p library `p-library-1-4` listed by the external data at `/shared/vhirschi/t3code/vakint_dev/p_integral_dev/wt/vk-bench4-xd` (commit `6eb49555`), shipped caches off, `130` measurements in `data/p_results.json`; cited: run `2026-11-08T18:42:02Z-84a7c0dee556`, gammaloop `a659e3bf8`, rustred `7c14cea0`, `1774` measurements.